1. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

Relevance: ★★★★★ (Highest)

- The solver must extract key information from the text:
  - Initial population: 10,000 penguins
  - Reproduction rule: one chick per couple
  - Mortality rate: 20%
- Natural language must be **translated into a mathematically manageable situation model**

👉 This is the **primary competence** required; misunderstanding at this stage invalidates all subsequent work.

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2. Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

Relevance: ★★★★★

- Recognising the situation as:
  - A population change problem
  - Involving both **increase (births)** and **decrease (deaths)**
- Identifying the **total number of penguins** as the key mathematical quantity of interest

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3. Identify the key variables in a model

Relevance: ★★★★☆

Key variables include:

- Initial number of penguins
- Number of penguin couples
- Number of chicks born
- Mortality rate
- Total population at the end of the year

👉 Even without symbolic notation, these variables must be **conceptually distinguished**.

4. Identify the constraints, assumptions and simplifications in a mathematical model

Relevance: ★★★★☆

The problem clearly states modelling assumptions:

- Each couple raises **exactly one chick**
- No migration, gender imbalance, or age-specific survival is considered
- The mortality rate applies **uniformly to all penguins**

👉 This reflects a **highly simplified (idealised) population model**.

5. Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

Relevance: ★★★★☆

The situation must be broken down into:

1. Initial adult population
2. Newly born chicks
3. Total population before deaths
4. Remaining population after mortality

👉 This decomposition is essential for converting a real-world context into a clear calculation process.

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6. Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures

Relevance: ★★★☆☆

- The task aligns with familiar:
  - Population growth models
  - Percentage decrease problems
- Can be compared to standard "increase then apply a reduction" scenarios

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7. Translate a problem into a standard mathematical representation or algorithm

Relevance: ★★★☆☆

- The situation naturally leads to an algorithm such as:

  > Final population = (initial population + number of chicks) × 0.8

- Even without formal algebraic notation, the reasoning is **algorithmic in nature**

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8. Recognise mathematical structure (including relationships and patterns) in problems or situations

Relevance: ★★★☆☆

- One-to-one relationship between couples and chicks
- Proportional relationship between mortality rate and total population

① Read, decode and make sense of statements… to create a model of the situation（理解题意并建立情境模型）

相关性：★★★★★（最高）

- 需要从文字中提取关键信息：
  - 初始数量：10,000 只
  - 繁殖规则：每对 1 只幼崽
  - 死亡率：20%
- 将自然语言**转化为一个可计算的情境模型**

👉 这是解题的**首要能力**，若理解错误，后续计算全部失效。

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② Identify and describe the mathematical aspects of a real-world problem situation（识别现实问题中的数学要素）

相关性：★★★★★

- 识别出这是一个：
  - 人口数量变化问题
  - 涉及**增长（出生）与减少（死亡）**
- 明确"企鹅总数"是最终关注的数学对象

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③ Identify the key variables in a model（识别模型中的关键变量）

相关性：★★★★☆

关键变量包括：

- 初始企鹅数量
- 企鹅情侣数量
- 新生幼崽数量
- 死亡比例
- 年末总数量

👉 虽然变量不需要用字母表示，但**概念上必须被区分清楚**

④ Identify the constraints, assumptions, simplifications in a mathematical model（识别模型的假设与简化）

相关性：★★★★☆

题目明确给出了建模假设：

- 每对企鹅**只养一只**幼崽
- 不考虑性别变化、迁移等因素
- 死亡率对**所有企鹅统一适用**

👉 这是**典型的"理想化人口模型"**

⑤ Simplify or decompose a situation to make it amenable to mathematical analysis（分解复杂情境）

相关性：★★★★☆

学生需要将问题拆解为：

1. 原有成体数量
2. 新增幼崽数量
3. 总数量
4. 死亡后的剩余数量

👉 这是将现实问题**转化为清晰计算步骤**的关键能力。

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⑥ Recognise aspects of a problem that correspond with known mathematical concepts or procedures（识别熟悉的数学结构）

相关性：★★★☆☆

- 属于常见的：
  - 人口增长模型
  - 百分比减少问题
- 可与"先加后乘折扣率"的问题类比

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⑦ Translate a problem into a standard mathematical representation or algorithm（转化为标准算法）

相关性：★★★☆☆

- 可自然形成计算流程：

  > 年末数量 =（原有数量 + 新生数量）× 0.8

- 虽未要求列代数式，但思路本质是**算法化的**

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⑧ Recognise mathematical structure (relationships and patterns)（识别数量关系）

相关性：★★★☆☆

- 情侣数量与幼崽数量的**一一对应关系**
- 死亡率与总数量的**比例关系**